

Pipeline Monitoring: Generic Dashboard: Mockup and Specification

Merck Data Observability

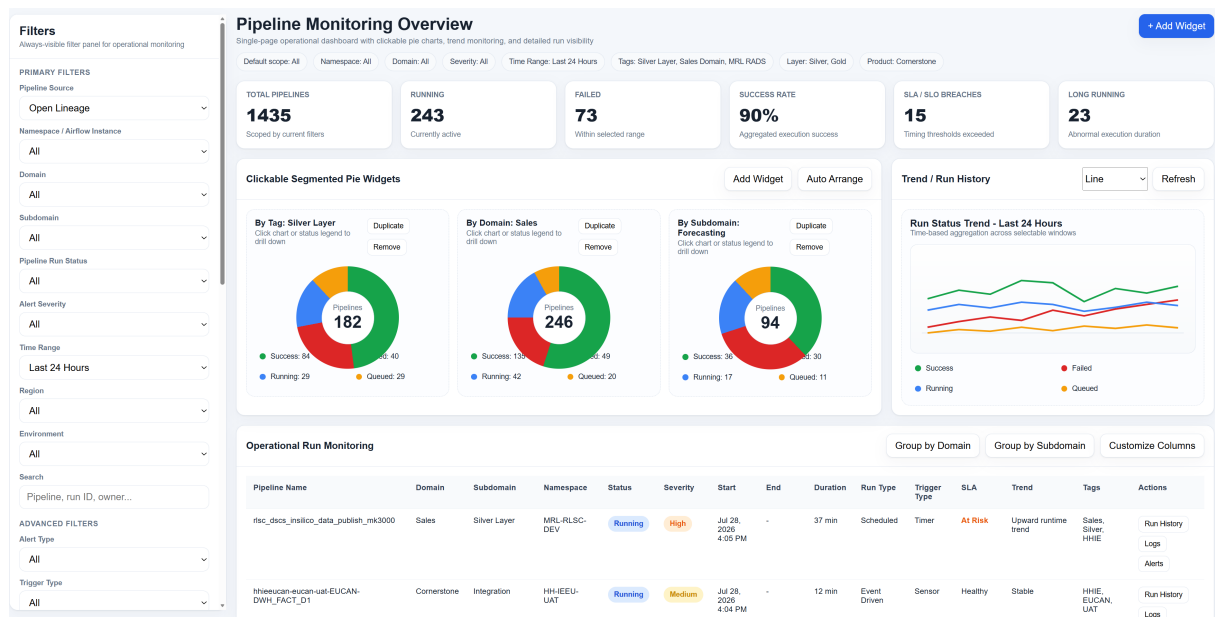
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Here is the interactive HTML

[pipe3.html](https://share.merck.com/download/attachments/2720644438/pipe3.html?api=v2&modificationDate=1785855030981&version=2)¹

1. <https://share.merck.com/download/attachments/2720644438/pipe3.html?api=v2&modificationDate=1785855030981&version=2>

1 Objectives

Objective	Description
Single operational view	Provide one page for day-to-day pipeline monitoring
Faster investigation	Enable users to narrow down issues using visible filters and drill-down widgets
Metadata-driven filtering	Support filtering by standard metadata and user-defined key-value filters
Better segmentation	Allow users to monitor by domain, subdomain, environment, region, team, tags, and similar dimensions
Configurable visualization	Allow users to add custom pie chart widgets for selected filter keys
Time-based monitoring	Support trend views across different time windows

2 Scope

2.1 In Scope

Item	Included
Sticky filter sidebar	Yes
KPI summary cards	Yes
Clickable pie widgets	Yes
Trend panel	Yes
Operational monitoring table	Yes
Multi-tag filtering	Yes
Custom key-value filters	Yes

2.2 Out of Scope

Item	Status
Alerts & Priority Watchlist section	Removed from final UI
Dependency Snapshot section	Removed from final UI
Separate tab-based pages	Not included
Free-text operator-based custom filters	Replaced by key + multiple values

3 Page Layout

Section	Description
Left Filter Sidebar	Always-visible filter section
Header Area	Dashboard title, subtitle, action buttons, active filter chips
KPI Summary	High-level health and run metrics
Pie Widget Area	Clickable user-configurable widgets
Trend Panel	Time-based run status visualization
Operational Run Monitoring Table	Detailed pipeline run information

4 Layout Behavior

Area	Behavior
Sidebar	Sticky on left side, independently scrollable if needed
Main content	Scrollable dashboard content
KPI cards	Responsive grid
Pie widgets	Responsive card grid
Trend panel	Displayed next to widget area on large screens
Run table	Full-width section below summary area

5 Filter Sidebar

5.1 Sidebar Behavior

Item	Specification
Position	Left side of the screen
Visibility	Always visible on desktop
Scroll behavior	Sidebar scrolls if filter content exceeds viewport height
Responsive behavior	Can stack on top for smaller screens

6 Primary Filters

Filter	Type	Values / Notes
Pipeline Source	Single select	Default value: Open Lineage
Namespace / Airflow Instance	Single select	All + known namespace values
Domain	Single select	All + configured domain values
Subdomain	Single select	All + configured subdomain values
Pipeline Run Status	Single select	All, Running, Failed, Success, Queued
Alert Severity	Single select	All, Critical, High, Medium, Low
Time Range	Single select	Last 10 Runs, Last 24 Hours, Last 3 Days, Last 7 Days, Weekly, Monthly
Region	Single select	All, APAC, EMEA, Americas
Environment	Single select	All, DEV, UAT, PROD
Search	Text input	Search by pipeline name, run ID, owner, or metadata text

7 Advanced Filters

Filter	Type	Values / Notes
Alert Type	Single select	All, Time, Status, Reliability, Missed Trigger, Long Running
Trigger Type	Single select	All, Timer, Sensor, Event Driven, Manual, Parent-Child
Run Type	Single select	All, Scheduled, Event Driven, Manual, Backfill
Criticality	Single select	All, Low, Medium, Major
Ownership Team	Single select	Team list including HHIE and MRL RADS
Trend Chart Mode	Single select	Line, Stacked Bar, Area, Heatmap

8 Multi-tag Filter

Item	Specification
Type	Multi-value tag input
Display	Values shown as chips
Add behavior	User enters a tag value and adds it
Remove behavior	Each tag chip can be removed
Scope behavior	Applied tags affect all dashboard sections

9 Custom Key-Value Filters

9.1 Functional Definition

Custom filters follow the Acceldata-style model:

- one **key**
- multiple **values** under that key

Example:

- Key = Layer
- Values = Silver , Gold

Example:

- Key = Product
- Values = Cornerstone , CSA-GAIN

9.2 Custom Filter UI Structure

Field	Type	Description
Filter Key	Single select	User selects metadata key
Values	Multi-value chips	User adds one or more values under the selected key
Add Value	Text input + action	Adds a value chip to the selected key
Remove Value	Chip action	Removes an individual value
Remove Filter	Card action	Removes the full key-value filter

9.3 Suggested Filter Keys

Filter Key	Description
Domain	Business domain
Subdomain	Domain subgroup
Product	Product or product family
Region	Geographic region
Environment	Deployment environment
Team	Team ownership
Tags	General metadata tags
Layer	Logical or medallion layer
Criticality	Business criticality
Trigger Type	Trigger classification
Run Type	Execution type
Owner	User or support owner
Business Unit	Business grouping

9.4 Expected Filter Logic

Rule	Recommended Behavior
Multiple values within one key	OR logic
Multiple different keys	AND logic

Example:

- Layer = Silver, Gold
- Product = Cornerstone

Recommended interpretation:

- include records where Layer is Silver or Gold
 - and Product is Cornerstone
-

10 Operational Flags

Flag	Type	Description
Long-running only	Checkbox	Show only long-running executions
SLA/SLO breaches only	Checkbox	Show only threshold breaches
Missed trigger only	Checkbox	Show only delayed or non-started expected executions
Queued child DAGs only	Checkbox	Reserved for future orchestration-linked filtering
Outliers only	Checkbox	Show only anomalous runs

11 Sidebar Actions

Action	Description
Apply	Apply all selected filters to the dashboard
Reset	Reset all filters to default values
Save View	Save current dashboard state for reuse

12 Header Area

Element	Description
Page Title	Pipeline Monitoring Overview
Subtitle	Short description of dashboard purpose
Action Button	Add Widget
Active Scope Chips	Display currently applied filters

13 Active Scope Chips

Item	Description
Purpose	Summarize applied filter context
Placement	Below page title
Content	Includes primary filters, tag filters, and custom key-value filters

13.1 Example Active Scope Chips

- Default scope: MRL RADS
 - Namespace: MRL-RLSC-UAT
 - Domain: Sales
 - Time Range: Last 24 Hours
 - Layer: Silver, Gold
 - Product: Cornerstone
-

14 KPI Summary

KPI	Description
Total Pipelines	Total number of pipelines in current scope
Running	Number of pipelines currently running
Failed	Number of failed pipelines in selected window
Success Rate	Aggregate success percentage
SLA / SLO Breaches	Number of threshold breaches
Long Running	Number of long-running pipelines

14.1 KPI Behavior

Behavior	Description
Refresh on filter apply	Yes
Respect all filters	Yes
Time range aware	Yes, where applicable

15 Pie Widget Area

15.1 Purpose

Provide segmented health and run status views by selected metadata dimensions.

15.2 Widget Components

Component	Description
Widget Title	User-defined label
Widget Subtitle	Explains breakdown logic
Pie Chart	Run status distribution
Center Count	Total volume in widget scope
Legend	Status counts
Widget Actions	Duplicate, Remove

15.3 Widget Configuration

Field	Type	Description
Widget Title	Text	User-defined widget label
Segment By	Single select	Tag, Domain, Subdomain, Namespace, Region, Environment, Criticality
Focus Value	Text or controlled value	Selected dimension value

Field	Type	Description
Measure	Single select	Pipeline Count, Run Count, Failed Run Count, SLA Breach Count

15.4 Widget Interactions

Interaction	Behavior
Click pie chart	Applies widget scope as dashboard filter
Click status legend item	Applies widget scope + status filter
Add Widget	Opens widget configuration
Duplicate	Duplicates selected widget
Remove	Removes widget
Auto Arrange	Reorders widget layout

15.5 Drill-down Examples

Trigger	Result
Click pie for Domain = Sales	Set Domain filter to Sales
Click pie for Subdomain = Forecasting	Set Subdomain filter to Forecasting
Click pie for Tag = Silver Layer	Add Silver Layer to tag filter
Click legend item Failed in Sales widget	Set Domain = Sales and Status = Failed

16 Trend Panel

16.1 Purpose

Provide time-based run visibility for the current dashboard scope.

16.2 Trend Controls

Control	Values
Time Range	Last 10 Runs, Last 24 Hours, Last 3 Days, Last 7 Days, Weekly, Monthly
Chart Mode	Line, Stacked Bar, Area, Heatmap
Refresh	Manual refresh action

16.3 Trend Output

Output	Description
Trend Title	Reflects selected time range
Series	Success, Failed, Running, Queued
Chart Type	Based on selected chart mode

17 Operational Run Monitoring Table

17.1 Purpose

Provide detailed execution-level information for operational follow-up.

17.2 Table Columns

Column	Description
Pipeline Name	Pipeline or DAG identifier
Domain	Business domain
Subdomain	Domain subgroup
Namespace	Airflow instance / namespace
Status	Run state
Severity	Alert severity or impact level
Start	Start timestamp
End	End timestamp
Duration	Runtime duration
Run Type	Scheduled, manual, backfill, event-driven
Trigger Type	Timer, sensor, parent-child, manual
SLA	SLA status
Trend	Short runtime / failure indicator

Column	Description
Tags	Associated metadata tags
Actions	Quick actions for investigation

17.3 Row Actions

Action	Description
Run History	Open run history
Logs	Open pipeline logs
Alerts	Open related alerts context

17.4 Table Behavior

Behavior	Description
Applies current filters	Yes
Supports grouping actions	Yes
Horizontal scroll	Yes, if needed
Highlight important rows	Yes, using status/severity styling

18 Personalization

Feature	Description
Save View	Save current filter + layout state
Default Scope	Allow team-specific or personal default scope
Widget Personalization	Allow user-defined widget setup

19 Default UI State

Item	Default Value
Pipeline Source	Open Lineage
Namespace	All
Domain	All
Subdomain	All
Run Status	All
Alert Severity	All
Time Range	Last 24 Hours
Region	All
Environment	All
Ownership Team	All or configured default
Trend Mode	Line

20 Assumptions

Assumption
Metadata is available and consistently maintained
Custom filter keys map to valid metadata fields
Time-based run metrics can be aggregated for the selected windows
Run metadata includes timestamps, status, tags, and execution type
Long-running and SLA definitions are handled outside the UI layer

21 Open Points

Topic	Clarification Needed
Long-running definition	Threshold logic by team/domain/environment
SLA/SLO logic	Exact calculation and breach rules
Criticality mapping	Standard values and business interpretation
Key-value filter semantics	Final backend behavior for OR/AND evaluation
Allowed metadata keys	Final governed list for custom filter keys
Search scope	Which fields are included in search

22 Requirement Alignment Summary

Requirement Area	UI Support
Single operational monitoring view	Supported
Domain / environment / region segmentation	Supported
Metadata-driven filtering	Supported
Multiple tag-based filtering	Supported
Custom filter keys with multiple values	Supported
Configurable pie chart views	Supported
Drill-down from pie chart	Supported
Time-range based trend view	Supported
Operational run detail view	Supported
